

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[HN_FFCutsFits,CutFitFileHN] =
recover_selected_cuts('HN_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multic
ut_allJ\sel_cuts');
```

```
fn = fieldnames(HN_FFCutsFits);
nf = 11
```

```
nf =
11
```

```
%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800HN_off1H1H0minFF'
```

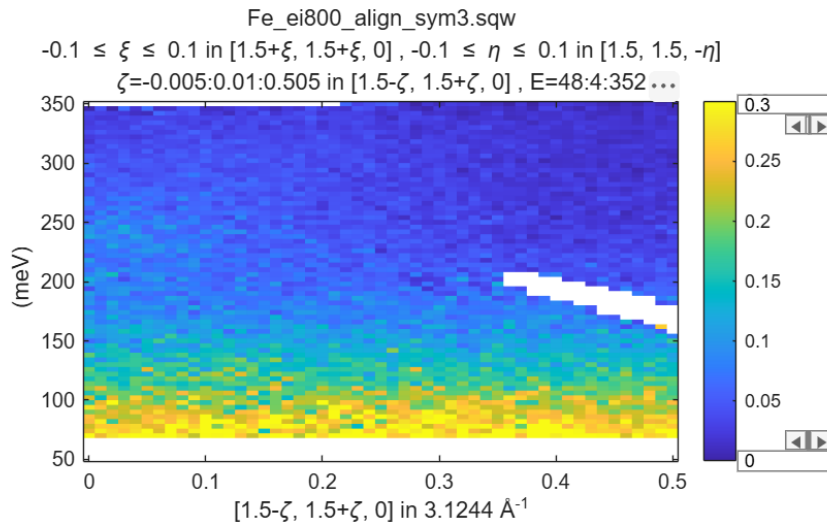
```
selectCuts =
'Ei800HN_off1H1H0minFF'
```

```
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end
```

```
fprintf('ProcessiHN cuts: %s \n', selectCuts);
```

```
ProcessiHN cuts: Ei800HN_off1H1H0minFF
```

```
cuts_list = HN_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
    cuts_list{j} = cuts_list{j}.set_instrument(mi);
    cuts_list{j} = cuts_list{j}.set_sample(sample);
    plot(cuts_list{j}); lz 0 0.3; keep_figure;
end
```



```
dir_name = "HN";
```

```
dE_step = 2; %original energy transfer step data were binned to. No point in going finer
```

```
half_dE = 5; % half width of data binning
```

```
en_mm = cuts_list{1}.data.img_range(:,4);
```

```
en_mm(1) = max(min_Ei,en_mm(1));
```

```
en_mm(1) = 200;
```

```
%en_mm(2) = 380;
```

```
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);
```

```
cut_dimensions = 1;
```

```
HN_FFcutsFits.(selectCuts) = cuts_list;
```

```
HN_FFcutsFits = fit_multicutsFF_along_direction(...
```

```
HN_FFcutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

```
*****
**** En = 205±5 going to 345
*****
```

```
Beginning fit (max 20 iterations)-----
```

```
Starting point
```

```
-----
Total time = 0.55114s    Reduced Chi^2 = 4.2494
```

```
Free parameter values:
```

```
10      0.7917      250      34      0.06011
```

```
-----
Iteration = 1
```

```
-----
Total time = 2.3189s    Reduced Chi^2 = 7.6566
```

```
Levenberg-Marquardt = 0.1
```

```
Free parameter values:
```

```
340.1    30.83    -5586    220.5    0.03735
```

```
Total time = 2.6411s    Reduced Chi^2 = 4.3467
```

```
Levenberg-Marquardt = 1
```

```
Free parameter values:
```

```
156      13.59    -3016    52.31    0.04853
```

```
Total time = 2.9659s    Reduced Chi^2 = 17.2453
```

```
Levenberg-Marquardt = 10
```

```
Free parameter values:
```

```
60.76    5.202    -911.8    34.3    0.05614
```

Total time = 3.2852s	Reduced Chi^2 = 4.0737	Levenberg-Marquardt = 1000
Free parameter values:		
15.18	1.241	131.4
		33.87
		0.05971

Iteration = 2

Total time = 4.8676s	Reduced Chi^2 = 3.6861	Levenberg-Marquardt = 100
Free parameter values:		
17.52	1.689	105.6
		32.28
		0.05847

Iteration = 3

Total time = 6.4191s	Reduced Chi^2 = 2.8014	Levenberg-Marquardt = 10
Free parameter values:		
23.78	2.129	93.5
		30.31
		0.05473

Iteration = 4

Total time = 8.023s	Reduced Chi^2 = 1.5834	Levenberg-Marquardt = 1
Free parameter values:		
29.92	2.419	84.47
		29.44
		0.04621

Iteration = 5

Total time = 9.5872s	Reduced Chi^2 = 1.2222	Levenberg-Marquardt = 0.1
Free parameter values:		
39.4	2.435	87.94
		29.2
		0.0392

Iteration = 6

Total time = 11.0832s	Reduced Chi^2 = 1.3795	Levenberg-Marquardt = 0.01
Free parameter values:		
81.41	2.286	95.91
		33.15
		0.03624
Total time = 11.384s	Reduced Chi^2 = 1.2175	Levenberg-Marquardt = 0.1
Free parameter values:		
53.51	2.404	90.45
		30.42
		0.03775

Iteration = 7

Total time = 12.9213s	Reduced Chi^2 = 1.8629	Levenberg-Marquardt = 0.01
Free parameter values:		
113.7	1.605	91.6
		33.91
		0.03563
Total time = 13.2384s	Reduced Chi^2 = 1.2187	Levenberg-Marquardt = 0.1
Free parameter values:		
73.6	2.166	91.02
		31.49
		0.037
Total time = 13.5295s	Reduced Chi^2 = 1.2022	Levenberg-Marquardt = 1
Free parameter values:		
60.15	2.338	90.35
		30.72
		0.03753

Iteration = 8

Total time = 15.0508s	Reduced Chi^2 = 1.1985	Levenberg-Marquardt = 0.1
Free parameter values:		
72.35	2.502	88.39
		32.7
		0.03701

Iteration = 9

Total time = 16.5868s	Reduced Chi^2 = 1.2409	Levenberg-Marquardt = 0.01
Free parameter values:		
103.6	2.1	88.72
		33.81
		0.03607
Total time = 16.8902s	Reduced Chi^2 = 1.2001	Levenberg-Marquardt = 0.1
Free parameter values:		
82.85	2.38	88.78
		33.04
		0.03667
Total time = 17.1908s	Reduced Chi^2 = 1.2001	Levenberg-Marquardt = 1

Free parameter values:

75.78	2.466	88.44	32.8	0.03691
Total time = 17.5114s	Reduced Chi^2 = 1.2019		Levenberg-Marquardt = 100	
Free parameter values:				
72.71	2.499	88.38	32.71	0.03701
Total time = 17.8278s	Reduced Chi^2 = 1.198		Levenberg-Marquardt = 10000	
Free parameter values:				
72.38	2.502	88.39	32.7	0.03701

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	72.38	+/-	534.4
4	2.502	+/-	52.05
5	88.39	+/-	716.8
6	32.7	+/-	330.1
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.03701	+/-	0.006022
2	0		
3	0		

Covariance matrix for free parameters:

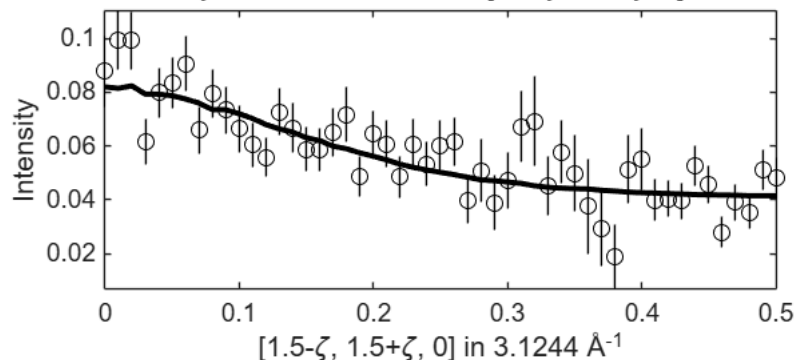
1.0000	0.5889	-0.9008	0.7806	0.1655
0.5889	1.0000	-0.8455	0.9629	0.3614
-0.9008	-0.8455	1.0000	-0.9565	-0.4405
0.7806	0.9629	-0.9565	1.0000	0.3812
0.1655	0.3614	-0.4405	0.3812	1.0000

Fe_ei800_align_sym3.sqw

S= 2.5± 52; J0= 33±3.3e+02; J1= 0± 0; gamma=72.3844±534.39

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $200 \leq$

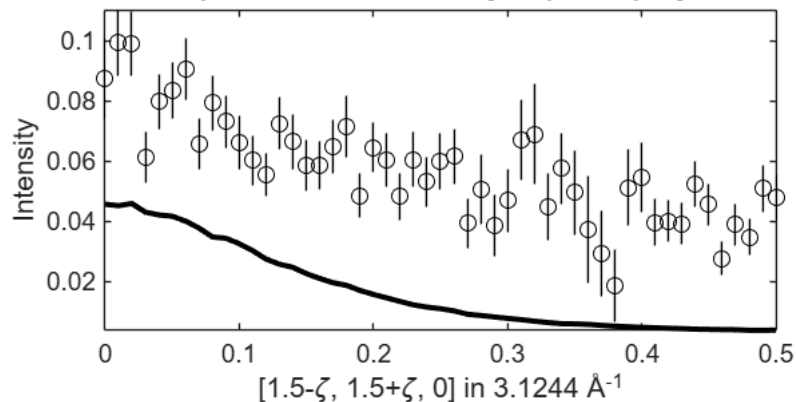
$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $200 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



**** En = 215±5 going to 345

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.36528s Reduced Chi^2 = 9.8484

Free parameter values:

72.38	2.502	88.39	32.7	0.0563
-------	-------	-------	------	--------

Iteration = 1

Total time = 2.1472s Reduced Chi^2 = 0.79416 Levenberg-Marquardt = 0.1

Free parameter values:

70.13	2.455	97.63	34.15	0.03996
-------	-------	-------	-------	---------

Iteration = 2

Total time = 3.9481s Reduced Chi^2 = 0.73704 Levenberg-Marquardt = 0.01

Free parameter values:

98.85	3.639	134.1	37.28	0.03633
-------	-------	-------	-------	---------

Iteration = 3

Total time = 5.631s Reduced Chi^2 = 0.75428 Levenberg-Marquardt = 0.001

Free parameter values:

166.3	4.227	209.2	36.26	0.03316
-------	-------	-------	-------	---------

Total time = 5.9906s Reduced Chi^2 = 0.72645 Levenberg-Marquardt = 0.01

Free parameter values:

130.9	3.899	169.9	36.54	0.03493
-------	-------	-------	-------	---------

Iteration = 4

Total time = 7.6939s Reduced Chi^2 = 0.80531 Levenberg-Marquardt = 0.001

Free parameter values:

251.9	4.883	217.8	44.92	0.03229
-------	-------	-------	-------	---------

Total time = 8.0563s Reduced Chi^2 = 0.72133 Levenberg-Marquardt = 0.01

Free parameter values:

179.1	4.223	193.4	38.71	0.03361
-------	-------	-------	-------	---------

Iteration = 5

Total time = 9.7777s Reduced Chi^2 = 1.201 Levenberg-Marquardt = 0.001

Free parameter values:

228.3	10.16	323.4	51.76	0.03131
Total time = 10.1402s		Reduced Chi^2 = 0.73063		Levenberg-Marquardt = 0.01
Free parameter values:				
202.5	6.095	239	42.45	0.03264
Total time = 10.4819s		Reduced Chi^2 = 0.71723		Levenberg-Marquardt = 0.1
Free parameter values:				
187.8	4.83	207.5	39.8	0.03333

Iteration = 6

Total time = 12.2445s		Reduced Chi^2 = 0.71517		Levenberg-Marquardt = 0.01
Free parameter values:				
223.5	5.822	237	42.71	0.03251

Iteration = 7

Total time = 13.9042s		Reduced Chi^2 = 1.1912		Levenberg-Marquardt = 0.001
Free parameter values:				
180.7	11.08	341.7	45.73	0.03102
Total time = 14.2334s		Reduced Chi^2 = 0.72209		Levenberg-Marquardt = 0.01
Free parameter values:				
213.3	7.481	272.6	43.43	0.03197
Total time = 14.5915s		Reduced Chi^2 = 0.71711		Levenberg-Marquardt = 0.1
Free parameter values:				
220.8	6.353	248.2	42.9	0.03238
Total time = 14.9129s		Reduced Chi^2 = 0.71655		Levenberg-Marquardt = 10
Free parameter values:				
223.5	5.88	237.9	42.71	0.03252
Total time = 15.2309s		Reduced Chi^2 = 0.71909		Levenberg-Marquardt = 1000
Free parameter values:				
223.5	5.828	237.1	42.71	0.03252
Total time = 15.578s		Reduced Chi^2 = 0.71934		Levenberg-Marquardt = 100000
Free parameter values:				
223.5	5.822	237	42.71	0.03251

*** No improvement in chi-squared over previous iteration ***

Total time = 15.5856s		Reduced Chi^2 = 0.71517		
Free parameter values:				
223.5	5.822	237	42.71	0.03251

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	223.5	+/-	3999
4	5.822	+/-	92.32
5	237	+/-	1616
6	42.71	+/-	177.8
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.03251	+/-	0.0214
2	0		
3	0		

Covariance matrix for free parameters:

```

-----
1.0000  -1.0000  -0.9947  0.9914  0.9399
-1.0000  1.0000  0.9951  -0.9919  -0.9415
-0.9947  0.9951  1.0000  -0.9996  -0.9687
0.9914  -0.9919  -0.9996  1.0000  0.9749
0.9399  -0.9415  -0.9687  0.9749  1.0000

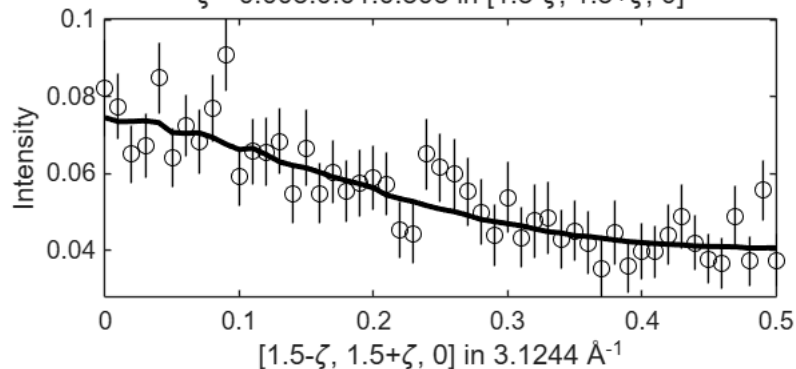
```

Fe_ei800_align_sym3.sqw

S= 5.8± 92; J0= 43±1.8e+02; J1= 0± 0; gamma=223.473±3999.3

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $210 \leq$

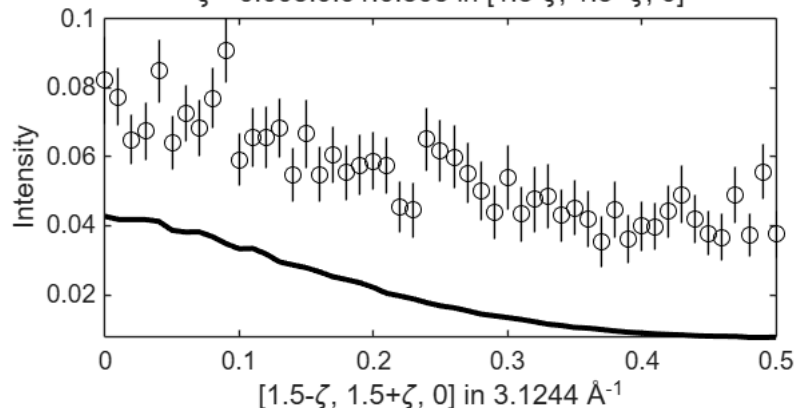
$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $210 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



```

*****
**** En = 225±5 going to 345
*****

```

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.36429s Reduced Chi^2 = 15.8132

Free parameter values:

223.5 5.822 237 42.71 0.05445

Iteration = 1

Total time = 2.2036s Reduced Chi^2 = 1.0353

Levenberg-Marquardt = 0.1

Free parameter values:

175.5 7.212 186.9 49.14 0.0288

Iteration = 2

```

Total time = 4.07s    Reduced Chi^2 = 1.0093    Levenberg-Marquardt = 0.01
Free parameter values:
    181.4        12.21        152.3        65.19        0.0262
-----
Iteration = 3
-----
Total time = 5.9442s    Reduced Chi^2 = 0.90977    Levenberg-Marquardt = 0.001
Free parameter values:
    243.5        20.55        78.24        98.02        0.0282
-----
Iteration = 4
-----
Total time = 7.7833s    Reduced Chi^2 = 21.5529    Levenberg-Marquardt = 0.0001
Free parameter values:
    1497        -9.865        -166.2        242.1        0.02938
Total time = 8.1669s    Reduced Chi^2 = 3.8584    Levenberg-Marquardt = 0.001
Free parameter values:
    634.5        10.61        -26.54        144.2        0.02869
Total time = 8.5384s    Reduced Chi^2 = 0.96234    Levenberg-Marquardt = 0.01
Free parameter values:
    367.8        17.54        34.9        112.4        0.02818
Total time = 8.9264s    Reduced Chi^2 = 0.85549    Levenberg-Marquardt = 1
Free parameter values:
    259.2        20.52        69.41        99.01        0.02815
-----
Iteration = 5
-----
Total time = 10.7245s    Reduced Chi^2 = 0.84583    Levenberg-Marquardt = 0.1
Free parameter values:
    300.1        20.12        55.7        104.6        0.02798
-----
Iteration = 6
-----
Total time = 12.6281s    Reduced Chi^2 = 0.8503    Levenberg-Marquardt = 0.01
Free parameter values:
    372.3        24.45        26.23        123.9        0.02829
Total time = 13.007s    Reduced Chi^2 = 0.84163    Levenberg-Marquardt = 0.1
Free parameter values:
    323.2        21.51        45.44        110.7        0.02807
-----
Iteration = 7
-----
Total time = 14.7855s    Reduced Chi^2 = 0.84565    Levenberg-Marquardt = 0.01
Free parameter values:
    373.2        28.19        17.51        131.4        0.0284
Total time = 15.1353s    Reduced Chi^2 = 0.85242    Levenberg-Marquardt = 0.1
Free parameter values:
    338.9        23.62        35.95        117.2        0.02816
Total time = 15.5061s    Reduced Chi^2 = 0.84893    Levenberg-Marquardt = 1
Free parameter values:
    328.3        22.19        42.29        112.8        0.02809
Total time = 15.8956s    Reduced Chi^2 = 0.84917    Levenberg-Marquardt = 100
Free parameter values:
    323.7        21.58        45.1        110.9        0.02807
Total time = 16.2913s    Reduced Chi^2 = 0.85254    Levenberg-Marquardt = 10000
Free parameter values:
    323.3        21.52        45.4        110.7        0.02807
Total time = 16.6688s    Reduced Chi^2 = 0.84825    Levenberg-Marquardt = 1000000
Free parameter values:
    323.2        21.51        45.43        110.7        0.02807
*** No improvement in chi-squared over previous iteration ***
Total time = 16.6771s    Reduced Chi^2 = 0.84163
Free parameter values:
    323.2        21.51        45.44        110.7        0.02807

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1 1
2 8
3 323.2 +/- 1.111e+04
4 21.51 +/- 165.5
5 45.44 +/- 1749
6 110.7 +/- 1174
7 0
8 0
9 0
10 0

Background parameter values:

1 0.02807 +/- 0.006811
2 0
3 0

Covariance matrix for free parameters:

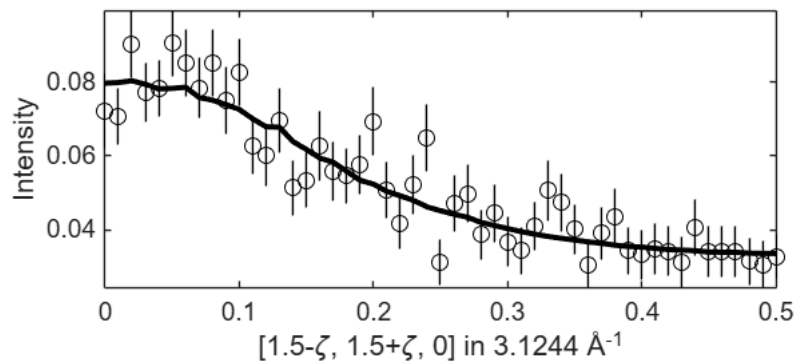
1.0000 -0.8979 -0.9917 0.9909 0.8043
-0.8979 1.0000 0.8338 -0.8304 -0.4828
-0.9917 0.8338 1.0000 -1.0000 -0.8681
0.9909 -0.8304 -1.0000 1.0000 0.8708
0.8043 -0.4828 -0.8681 0.8708 1.0000

Fe_ei800_align_sym3.sqw

22±1.7e+02; J0=1.1e+02±1.2e+03; J1= 0± 0; gamma=323.201±11

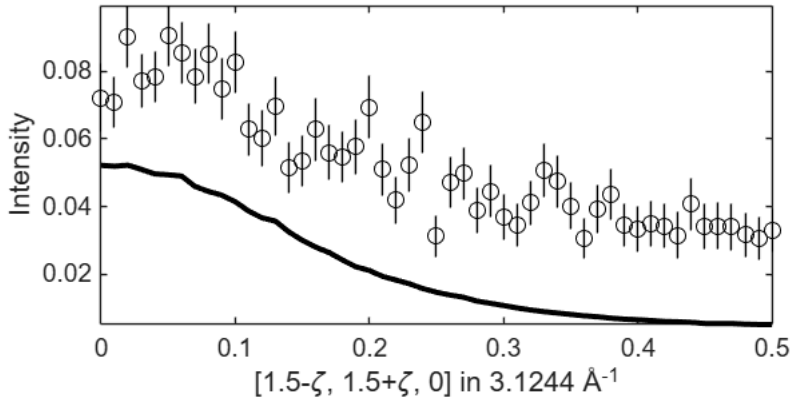
$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $220 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $220 \leq \zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



 **** En = 235±5 going to 345

Beginning fit (max 20 iterations)-----
 Starting point

Total time = 0.45996s Reduced Chi^2 = 19.1915
 Free parameter values:
 323.2 21.51 45.44 110.7 0.05145

Iteration = 1

Total time = 2.6004s Reduced Chi^2 = 0.94216 Levenberg-Marquardt = 0.1
 Free parameter values:
 235.8 21.95 22.54 105.8 0.02546

Iteration = 2

Total time = 4.7454s Reduced Chi^2 = 0.85411 Levenberg-Marquardt = 0.01
 Free parameter values:
 127.5 18.51 58.45 76.82 0.0224

Iteration = 3

Total time = 6.8517s Reduced Chi^2 = 6.5103 Levenberg-Marquardt = 0.001
 Free parameter values:
 25.6 2.003 141.3 24.92 0.02177
 Total time = 7.2724s Reduced Chi^2 = 0.83577 Levenberg-Marquardt = 0.01
 Free parameter values:
 96.83 13.48 86.06 60.2 0.02218

Iteration = 4

Total time = 9.4173s Reduced Chi^2 = 4.2519 Levenberg-Marquardt = 0.001
 Free parameter values:
 -13.03 5.742 162.7 18.58 0.02163
 Total time = 9.876s Reduced Chi^2 = 0.84127 Levenberg-Marquardt = 0.01
 Free parameter values:
 62.3 11.07 110.5 46.98 0.02199
 Total time = 10.2843s Reduced Chi^2 = 0.82067 Levenberg-Marquardt = 0.1
 Free parameter values:
 86.06 12.74 93.56 55.99 0.02213

Iteration = 5

```

-----
Total time = 12.394s   Reduced Chi^2 = 1.1046   Levenberg-Marquardt = 0.01
Free parameter values:
    43.16      12.98      116.3      44.98      0.02201
Total time = 12.8492s   Reduced Chi^2 = 0.82524   Levenberg-Marquardt = 0.1
Free parameter values:
    72.56      12.83      100.6      52.49      0.02208
Total time = 13.2867s   Reduced Chi^2 = 0.82987   Levenberg-Marquardt = 1
Free parameter values:
    81.86      12.78      95.66      54.87      0.02212
Total time = 13.8201s   Reduced Chi^2 = 0.83869   Levenberg-Marquardt = 100
Free parameter values:
    85.65      12.74      93.75      55.88      0.02213
Total time = 14.263s   Reduced Chi^2 = 0.83113   Levenberg-Marquardt = 10000
Free parameter values:
    86.02      12.74      93.58      55.98      0.02213
Total time = 14.6752s   Reduced Chi^2 = 0.83909   Levenberg-Marquardt = 1000000
Free parameter values:
    86.06      12.74      93.56      55.99      0.02213
*** No improvement in chi-squared over previous iteration ***
Total time = 14.6825s   Reduced Chi^2 = 0.82067
Free parameter values:
    86.06      12.74      93.56      55.99      0.02213
-----

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
1          1
2          8
3          86.06 +/- 765.2
4          12.74 +/- 96.13
5          93.56 +/- 232.7
6          55.99 +/- 48.85
7          0
8          0
9          0
10         0

```

Background parameter values:

```

-----
1          0.02213 +/- 0.003975
2          0
3          0

```

Covariance matrix for free parameters:

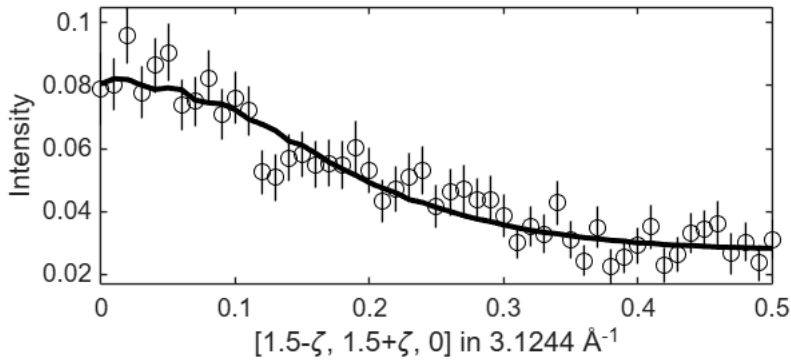
```

-----
1.0000 -0.9958 -0.8275 0.7863 -0.0260
-0.9958 1.0000 0.7726 -0.7265 0.1087
-0.8275 0.7726 1.0000 -0.9975 -0.5036
0.7863 -0.7265 -0.9975 1.0000 0.5548
-0.0260 0.1087 -0.5036 0.5548 1.0000

```

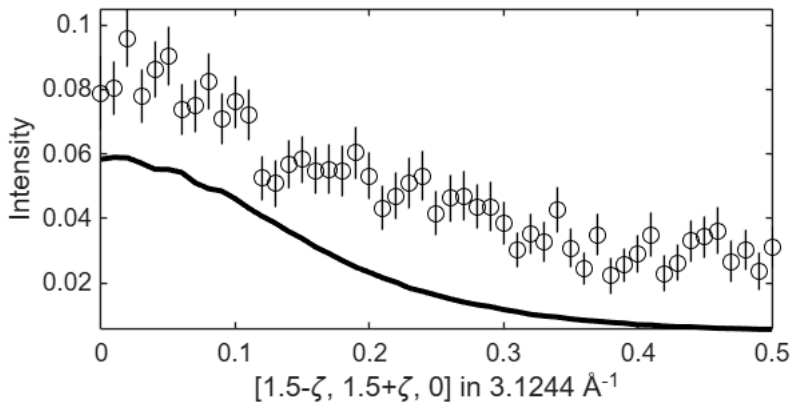
Fe_ei800_align_sym3.sqw

S= 13± 96; J0= 56± 49; J1= 0± 0; gamma=86.0612±765.155
 $\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $230 \leq$
 $\zeta=-0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $230 \leq$
 $\zeta=-0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



**** En = 245±5 going to 345

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.47081s Reduced Chi^2 = 27.1145

Free parameter values:

86.06	12.74	93.56	55.99	0.04801
-------	-------	-------	-------	---------

Iteration = 1

Total time = 2.6883s Reduced Chi^2 = 0.88694 Levenberg-Marquardt = 0.1

Free parameter values:

77.89	10.86	82.19	56.22	0.02275
-------	-------	-------	-------	---------

Iteration = 2

Total time = 5.0757s Reduced Chi^2 = 0.81791 Levenberg-Marquardt = 0.01

Free parameter values:

86.61	8.278	78.65	53.12	0.02024
-------	-------	-------	-------	---------

Iteration = 3

Total time = 7.4144s Reduced Chi^2 = 14.9468 Levenberg-Marquardt = 0.001

Free parameter values:

```

          99.77      0.4162      102.5      36.59      0.02006
Total time = 7.9041s   Reduced Chi^2 = 0.83009   Levenberg-Marquardt = 0.01
Free parameter values:
          91.24      5.83      85.09      47.79      0.02017
Total time = 8.4075s   Reduced Chi^2 = 0.79455   Levenberg-Marquardt = 0.1
Free parameter values:
          88.59      7.55      79.81      51.34      0.02021
-----
Iteration = 4
-----
Total time = 10.8091s   Reduced Chi^2 = 1.1658   Levenberg-Marquardt = 0.01
Free parameter values:
          110.2      4.444      75.69      48.8      0.02024
Total time = 11.3074s   Reduced Chi^2 = 0.80612   Levenberg-Marquardt = 0.1
Free parameter values:
          95.44      6.568      78.53      50.54      0.02023
Total time = 11.8255s   Reduced Chi^2 = 0.79341   Levenberg-Marquardt = 1
Free parameter values:
          90.77      7.24      79.4      51.09      0.02022
-----
Iteration = 5
-----
Total time = 14.1611s   Reduced Chi^2 = 0.80136   Levenberg-Marquardt = 0.1
Free parameter values:
          86.5      7.362      82.5      50.32      0.02022
Total time = 14.6317s   Reduced Chi^2 = 0.79848   Levenberg-Marquardt = 1
Free parameter values:
          89.43      7.279      80.37      50.84      0.02023
Total time = 15.1534s   Reduced Chi^2 = 0.80377   Levenberg-Marquardt = 10
Free parameter values:
          90.35      7.253      79.7      51.01      0.02023
Total time = 15.6234s   Reduced Chi^2 = 0.79814   Levenberg-Marquardt = 1000
Free parameter values:
          90.72      7.241      79.43      51.08      0.02023
Total time = 16.0842s   Reduced Chi^2 = 0.80384   Levenberg-Marquardt = 100000
Free parameter values:
          90.76      7.24      79.4      51.09      0.02022
Total time = 16.5257s   Reduced Chi^2 = 0.7956   Levenberg-Marquardt = 10000000
Free parameter values:
          90.77      7.24      79.4      51.09      0.02022
*** No improvement in chi-squared over previous iteration ***
Total time = 16.5345s   Reduced Chi^2 = 0.79341
Free parameter values:
          90.77      7.24      79.4      51.09      0.02022
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----
Foreground parameter values:
-----
  1          1
  2          8
  3      90.77 +/- 733.7
  4       7.24 +/- 27.29
  5       79.4 +/- 558.4
  6      51.09 +/- 122
  7          0
  8          0
  9          0
 10          0

```

Background parameter values:

```
-----
1      0.02022  +/-  0.004957
2      0
3      0
```

Covariance matrix for free parameters:

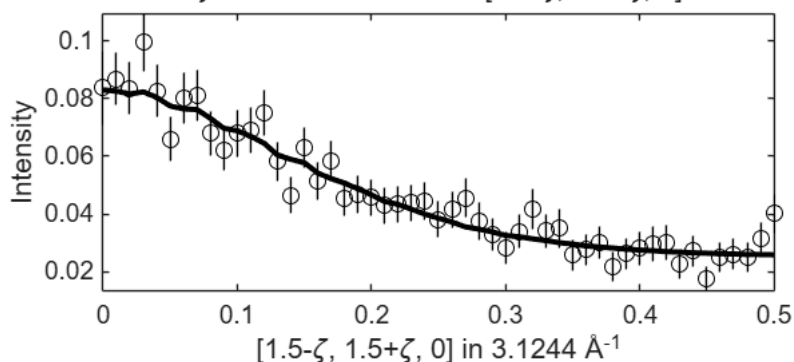
```
-----
1.0000  -0.9832  -0.9828   0.9800   0.7891
-0.9832   1.0000   0.9327  -0.9273  -0.6774
-0.9828   0.9327   1.0000  -0.9999  -0.8792
0.9800  -0.9273  -0.9999   1.0000   0.8844
0.7891  -0.6774  -0.8792   0.8844   1.0000
```

Fe_ei800_align_sym3.sqw

S= 7.2± 27; J0= 51±1.2e+02; J1= 0± 0; gamma=90.7657±733.74

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $240 \leq$

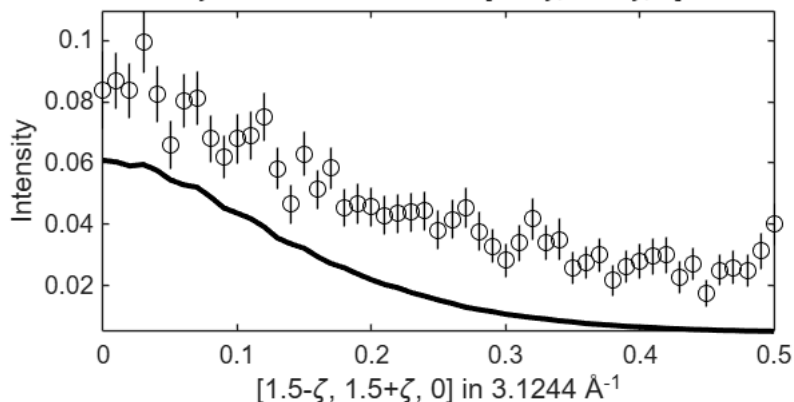
$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $240 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



**** En = 255±5 going to 345

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.48283s Reduced Chi^2 = 28.4952

Free parameter values:

90.77 7.24 79.4 51.09 0.04417

Iteration = 1

Total time = 2.935s	Reduced Chi^2 = 1.5426	Levenberg-Marquardt = 0.1
Free parameter values:		
83.89	7.626	102.2
		51.09
		0.02048

Iteration = 2

Total time = 5.3128s	Reduced Chi^2 = 1.3785	Levenberg-Marquardt = 0.01
Free parameter values:		
100.8	8.483	196.4
		42.01
		0.0151

Iteration = 3

Total time = 7.6231s	Reduced Chi^2 = 1.405	Levenberg-Marquardt = 0.001
Free parameter values:		
150.6	12.63	287.2
		44.75
		0.009606
Total time = 8.0834s	Reduced Chi^2 = 1.3425	Levenberg-Marquardt = 0.01
Free parameter values:		
125.8	10.07	242.3
		42.11
		0.01183

Iteration = 4

Total time = 10.4239s	Reduced Chi^2 = 34.0091	Levenberg-Marquardt = 0.001
Free parameter values:		
3.573	21.41	357.6
		31.08
		0.007415
Total time = 10.8843s	Reduced Chi^2 = 1.5853	Levenberg-Marquardt = 0.01
Free parameter values:		
93.17	13.79	290.1
		37.92
		0.009774
Total time = 11.3785s	Reduced Chi^2 = 1.3266	Levenberg-Marquardt = 0.1
Free parameter values:		
116.2	11.28	257.8
		40.7
		0.01112

Iteration = 5

Total time = 13.7524s	Reduced Chi^2 = 1.6132	Levenberg-Marquardt = 0.01
Free parameter values:		
86.45	15.73	298.6
		37.99
		0.009347
Total time = 14.2496s	Reduced Chi^2 = 1.3297	Levenberg-Marquardt = 0.1
Free parameter values:		
107.3	12.71	271.1
		39.77
		0.01052
Total time = 14.7149s	Reduced Chi^2 = 1.3167	Levenberg-Marquardt = 1
Free parameter values:		
113.5	11.75	261.8
		40.38
		0.01095

Iteration = 6

Total time = 17.0676s	Reduced Chi^2 = 1.3205	Levenberg-Marquardt = 0.1
Free parameter values:		
107.4	13.74	276.3
		40.7
		0.01034
Total time = 17.6015s	Reduced Chi^2 = 1.3189	Levenberg-Marquardt = 1
Free parameter values:		
111.7	12.38	266.3
		40.47
		0.01075
Total time = 18.0718s	Reduced Chi^2 = 1.3205	Levenberg-Marquardt = 10
Free parameter values:		
113	11.95	263.2
		40.41
		0.01089
Total time = 18.5389s	Reduced Chi^2 = 1.3154	Levenberg-Marquardt = 1000
Free parameter values:		
113.5	11.77	261.9
		40.38
		0.01094

Iteration = 7

Total time = 20.8219s	Reduced Chi^2 = 1.3244	Levenberg-Marquardt = 100
Free parameter values:		
113.2	11.81	262.3
		40.36
		0.01092
Total time = 21.2843s	Reduced Chi^2 = 1.3193	Levenberg-Marquardt = 1000

```

Free parameter values:
  113.4      11.78      262.1      40.37      0.01094
Total time = 21.7491s   Reduced Chi^2 = 1.3323   Levenberg-Marquardt = 10000
Free parameter values:
  113.5      11.77      262      40.38      0.01094
Total time = 22.2121s   Reduced Chi^2 = 1.3189   Levenberg-Marquardt = 1000000
Free parameter values:
  113.5      11.77      261.9      40.38      0.01094
Total time = 22.6883s   Reduced Chi^2 = 1.3315   Levenberg-Marquardt = 100000000
Free parameter values:
  113.5      11.77      261.9      40.38      0.01094
Total time = 23.1643s   Reduced Chi^2 = 1.3212   Levenberg-Marquardt = 10000000000
Free parameter values:
  113.5      11.77      261.9      40.38      0.01094
*** No improvement in chi-squared over previous iteration ***
Total time = 23.1734s   Reduced Chi^2 = 1.3154
Free parameter values:
  113.5      11.77      261.9      40.38      0.01094

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
  1      1
  2      8
  3      113.5 +/- 3441
  4      11.77 +/- 355
  5      261.9 +/- 1076
  6      40.38 +/- 93.16
  7      0
  8      0
  9      0
 10      0

```

Background parameter values:

```

-----
  1      0.01094 +/- 0.01884
  2      0
  3      0

```

Covariance matrix for free parameters:

```

-----
  1.0000 -0.9999 -0.9954  0.9959  0.9512
 -0.9999  1.0000  0.9964 -0.9968 -0.9546
 -0.9954  0.9964  1.0000 -0.9999 -0.9751
  0.9959 -0.9968 -0.9999  1.0000  0.9723
  0.9512 -0.9546 -0.9751  0.9723  1.0000

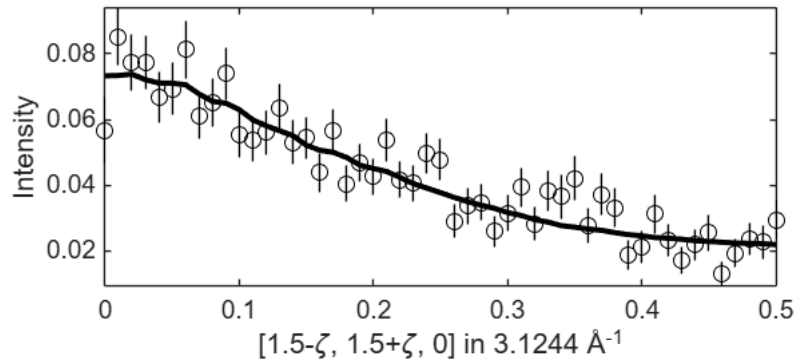
```

Fe_ei800_align_sym3.sqw

S= 12±3.5e+02; J0= 40± 93; J1= 0± 0; gamma=113.487±3441.1.

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $250 \leq$

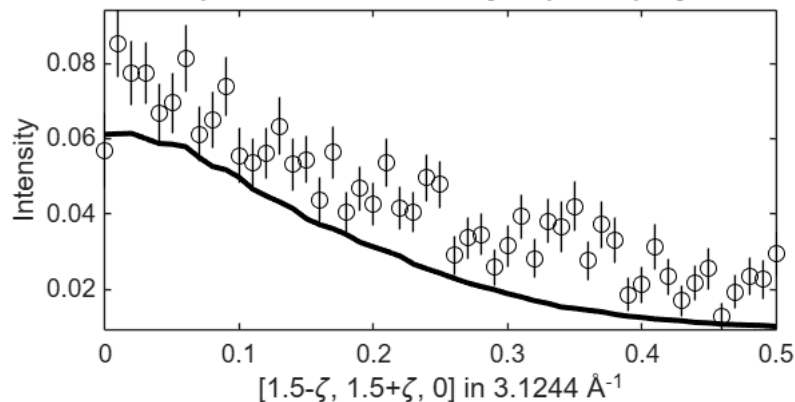
$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $250 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



**** En = 265±5 going to 345

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.53052s Reduced Chi^2 = 48.8401

Free parameter values:

113.5	11.77	261.9	40.38	0.04021
-------	-------	-------	-------	---------

Iteration = 1

Total time = 3.1714s Reduced Chi^2 = 1.785 Levenberg-Marquardt = 0.1

Free parameter values:

93.5	12.17	267.2	43.01	0.01568
------	-------	-------	-------	---------

Iteration = 2

Total time = 5.7722s Reduced Chi^2 = 1.595 Levenberg-Marquardt = 0.01

Free parameter values:

123.4	11.31	298.7	41.39	0.01056
-------	-------	-------	-------	---------

Iteration = 3

Total time = 8.1771s Reduced Chi^2 = 4.2216 Levenberg-Marquardt = 0.001

Free parameter values:

84.14	22.57	389.5	42.93	0.007414
Total time = 8.6997s	Reduced Chi^2 = 1.6057		Levenberg-Marquardt = 0.01	
Free parameter values:				
114.9	15.04	334	41.17	0.008973
Total time = 9.2174s	Reduced Chi^2 = 1.563		Levenberg-Marquardt = 0.1	
Free parameter values:				
121.7	12.57	308.7	41.17	0.009922

Iteration = 4

Total time = 11.7972s	Reduced Chi^2 = 1.5544		Levenberg-Marquardt = 0.01	
Free parameter values:				
108.9	14.39	333.1	38.53	0.00863

Iteration = 5

Total time = 14.3688s	Reduced Chi^2 = 2.2704		Levenberg-Marquardt = 0.001	
Free parameter values:				
98.06	25.78	400.1	44.22	0.006848
Total time = 14.8942s	Reduced Chi^2 = 1.5631		Levenberg-Marquardt = 0.01	
Free parameter values:				
106.9	18.1	357.9	39.94	0.007763
Total time = 15.4183s	Reduced Chi^2 = 1.5596		Levenberg-Marquardt = 0.1	
Free parameter values:				
108.7	15.61	340.5	38.91	0.008312
Total time = 15.9366s	Reduced Chi^2 = 1.5488		Levenberg-Marquardt = 10	
Free parameter values:				
109.1	14.54	333.5	38.53	0.008626

Iteration = 6

Total time = 18.3948s	Reduced Chi^2 = 1.5495		Levenberg-Marquardt = 1	
Free parameter values:				
108.9	14.95	335.6	38.67	0.008511
Total time = 18.901s	Reduced Chi^2 = 1.5516		Levenberg-Marquardt = 10	
Free parameter values:				
109.1	14.67	334.1	38.57	0.008598
Total time = 19.4276s	Reduced Chi^2 = 1.5498		Levenberg-Marquardt = 100	
Free parameter values:				
109.1	14.58	333.7	38.54	0.008618
Total time = 19.9917s	Reduced Chi^2 = 1.5503		Levenberg-Marquardt = 10000	
Free parameter values:				
109.1	14.54	333.5	38.54	0.008625
Total time = 20.4912s	Reduced Chi^2 = 1.5539		Levenberg-Marquardt = 1000000	
Free parameter values:				
109.1	14.54	333.5	38.54	0.008626
Total time = 20.9846s	Reduced Chi^2 = 1.5581		Levenberg-Marquardt = 100000000	
Free parameter values:				
109.1	14.54	333.5	38.53	0.008626

*** No improvement in chi-squared over previous iteration ***

Total time = 20.9924s	Reduced Chi^2 = 1.5488			
Free parameter values:				
109.1	14.54	333.5	38.53	0.008626

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1			
2	8			
3	109.1	+/-	4961	

4	14.54	+/-	674.5
5	333.5	+/-	1221
6	38.53	+/-	84.09
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.008626	+/-	0.02175
2	0		
3	0		

Covariance matrix for free parameters:

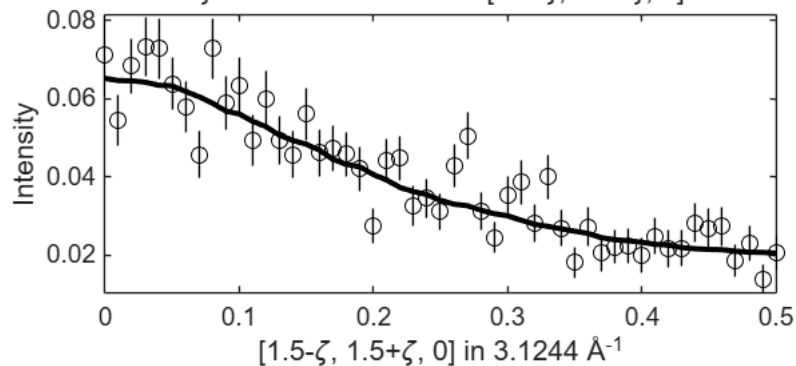
1.0000	-0.9999	-0.9940	0.9963	0.9467
-0.9999	1.0000	0.9953	-0.9972	-0.9507
-0.9940	0.9953	1.0000	-0.9994	-0.9752
0.9963	-0.9972	-0.9994	1.0000	0.9674
0.9467	-0.9507	-0.9752	0.9674	1.0000

Fe_ei800_align_sym3.sqw

S= 15±6.7e+02; J0= 39± 84; J1= 0± 0; gamma=109.074±4960.6

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $260 \leq$

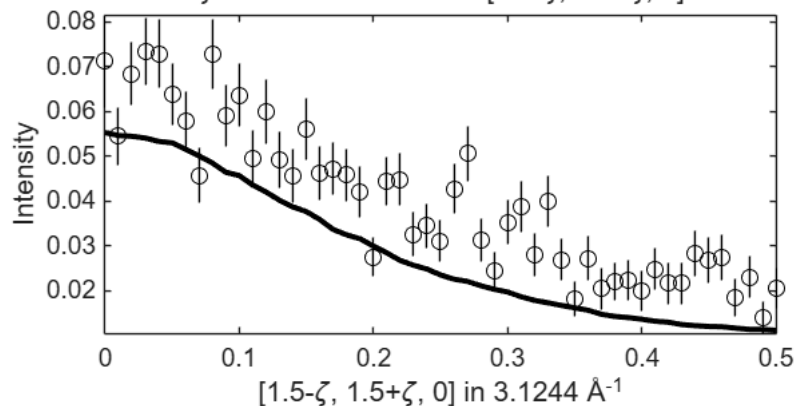
$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $260 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



**** En = 275±5 going to 345

```

Beginning fit (max 20 iterations)-----
Starting point
-----
Total time = 0.53267s    Reduced Chi^2 = 45.7997
Free parameter values:
    109.1          14.54          333.5          38.53          0.03864
-----
Iteration = 1
-----
Total time = 3.1604s    Reduced Chi^2 = 1.1373    Levenberg-Marquardt = 0.1
Free parameter values:
    106.8          13.48          328.3          42.46          0.015
-----
Iteration = 2
-----
Total time = 5.8625s    Reduced Chi^2 = 0.91179    Levenberg-Marquardt = 0.01
Free parameter values:
    142.3          12.52          357.5          41.19          0.009503
-----
Iteration = 3
-----
Total time = 8.5658s    Reduced Chi^2 = 1.6229    Levenberg-Marquardt = 0.001
Free parameter values:
    267.1          12.13          427.5          49.04          0.00709
Total time = 9.1013s    Reduced Chi^2 = 0.89303    Levenberg-Marquardt = 0.01
Free parameter values:
    185           12.54          384.2          42.99          0.008234
-----
Iteration = 4
-----
Total time = 11.8361s   Reduced Chi^2 = 1.3511    Levenberg-Marquardt = 0.001
Free parameter values:
    298.7          10.66          425.8          47.06          0.006696
Total time = 12.391s   Reduced Chi^2 = 0.87874    Levenberg-Marquardt = 0.01
Free parameter values:
    223.8          12.02          400.7          43.76          0.007502
-----
Iteration = 5
-----
Total time = 15.1778s   Reduced Chi^2 = 1.3078    Levenberg-Marquardt = 0.001
Free parameter values:
    297.6          8.249          397.6          42.55          0.006599
Total time = 15.6972s   Reduced Chi^2 = 0.87096    Levenberg-Marquardt = 0.01
Free parameter values:
    249.3          10.86          401.7          43.05          0.007056
-----
Iteration = 6
-----
Total time = 18.3902s   Reduced Chi^2 = 1.7094    Levenberg-Marquardt = 0.001
Free parameter values:
    441.7          8.969          425.2          53.41          0.006792
Total time = 18.934s   Reduced Chi^2 = 0.88401    Levenberg-Marquardt = 0.01
Free parameter values:
    312.3          10.28          410.8          46.06          0.006822
Total time = 19.4782s   Reduced Chi^2 = 0.86288    Levenberg-Marquardt = 0.1
Free parameter values:
    269.8          10.7          404.2          43.95          0.006977
-----
Iteration = 7
-----
Total time = 22.2476s   Reduced Chi^2 = 0.87765    Levenberg-Marquardt = 0.01
Free parameter values:
    337.1          10.57          417.5          47.86          0.006781
Total time = 22.7707s   Reduced Chi^2 = 0.86069    Levenberg-Marquardt = 0.1

```

```

Free parameter values:
    291.4      10.66      408.3      45.16      0.00689
-----
Iteration = 8
-----
Total time = 25.5565s    Reduced Chi^2 = 0.87214    Levenberg-Marquardt = 0.01
Free parameter values:
    359.9      10.71      422.3      49.32      0.00682
Total time = 26.1249s    Reduced Chi^2 = 0.86746    Levenberg-Marquardt = 0.1
Free parameter values:
    313.2      10.68      412.7      46.46      0.006875
Total time = 26.6783s    Reduced Chi^2 = 0.8631    Levenberg-Marquardt = 1
Free parameter values:
    298.4      10.67      409.6      45.56      0.006895
Total time = 27.208s    Reduced Chi^2 = 0.86785    Levenberg-Marquardt = 100
Free parameter values:
    292.1      10.67      408.4      45.2      0.006892
Total time = 27.7407s    Reduced Chi^2 = 0.86136    Levenberg-Marquardt = 10000
Free parameter values:
    291.4      10.66      408.3      45.17      0.00689
Total time = 28.2834s    Reduced Chi^2 = 0.86956    Levenberg-Marquardt = 1000000
Free parameter values:
    291.4      10.66      408.3      45.16      0.00689
*** No improvement in chi-squared over previous iteration ***
Total time = 28.2924s    Reduced Chi^2 = 0.86069
Free parameter values:
    291.4      10.66      408.3      45.16      0.00689
-----

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
1          1
2          8
3      291.4 +/- 2740
4      10.66 +/- 115.2
5      408.3 +/- 1212
6      45.16 +/- 70.04
7          0
8          0
9          0
10         0

```

Background parameter values:

```

-----
1      0.00689 +/- 0.02027
2          0
3          0

```

Covariance matrix for free parameters:

```

-----
1.0000  -0.9979  -0.9875   0.9949   0.9380
-0.9979   1.0000   0.9956  -0.9988  -0.9585
-0.9875   0.9956   1.0000  -0.9978  -0.9803
0.9949  -0.9988  -0.9978   1.0000   0.9652
0.9380  -0.9585  -0.9803   0.9652   1.0000

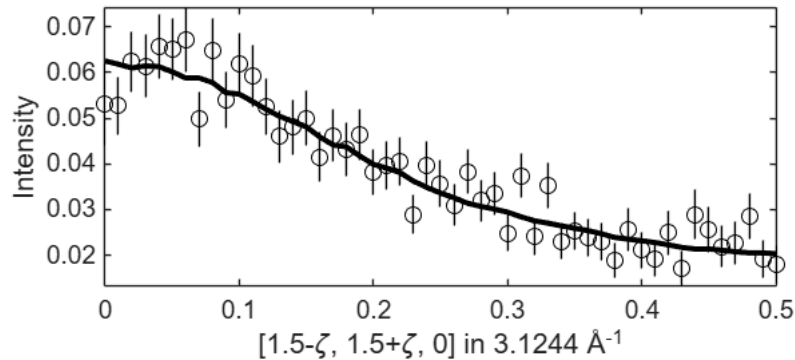
```

Fe_ei800_align_sym3.sqw

S= 11±1.2e+02; J0= 45± 70; J1= 0± 0; gamma=291.357±2740.4

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $270 \leq$

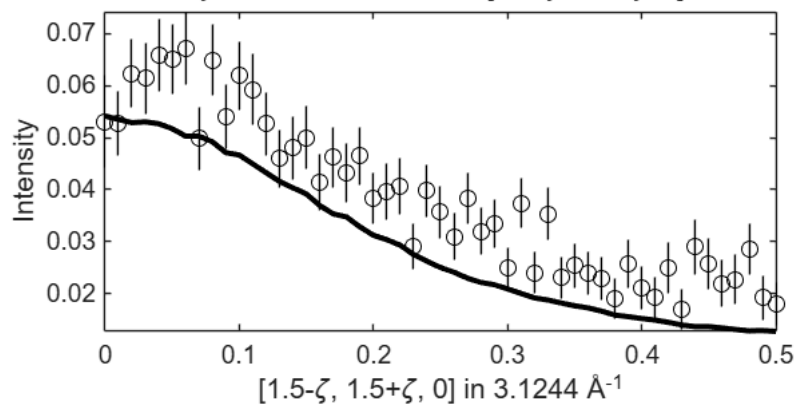
$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $270 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



**** En = 285±5 going to 345

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.61982s Reduced Chi^2 = 62.5209

Free parameter values:

291.4	10.66	408.3	45.16	0.03518
-------	-------	-------	-------	---------

Iteration = 1

Total time = 3.7743s Reduced Chi^2 = 1.7493 Levenberg-Marquardt = 0.1

Free parameter values:

297.9	8.305	413.2	48.6	0.01311
-------	-------	-------	------	---------

Iteration = 2

Total time = 6.8559s Reduced Chi^2 = 7.9945 Levenberg-Marquardt = 0.01

Free parameter values:

588.3	4.003	480	42.72	0.004819
-------	-------	-----	-------	----------

Total time = 7.4749s Reduced Chi^2 = 1.5195 Levenberg-Marquardt = 0.1

Free parameter values:

392.3	7.025	428.4	46.72	0.008533
-------	-------	-------	-------	----------

Iteration = 3

```

-----
Total time = 10.7617s    Reduced Chi^2 = 12.4305    Levenberg-Marquardt = 0.01
Free parameter values:
    672          2.42          395.6          39.6          0.003447
Total time = 11.4109s    Reduced Chi^2 = 1.4825    Levenberg-Marquardt = 0.1
Free parameter values:
    489.5        5.655          413.8          44.03          0.006333
-----
Iteration = 4
-----
Total time = 14.5543s    Reduced Chi^2 = 11.0427    Levenberg-Marquardt = 0.01
Free parameter values:
    695.9        2.142          276.6          40.45          0.00225
Total time = 15.1951s    Reduced Chi^2 = 1.4218    Levenberg-Marquardt = 0.1
Free parameter values:
    567.7        4.612          365           42.51          0.004434
-----
Iteration = 5
-----
Total time = 18.3444s    Reduced Chi^2 = 2.9359    Levenberg-Marquardt = 0.01
Free parameter values:
    714.8        3.171          188.2          49.13          0.002368
Total time = 18.9595s    Reduced Chi^2 = 1.3191    Levenberg-Marquardt = 0.1
Free parameter values:
    619.6        4.238          300.6          44.37          0.003085
-----
Iteration = 6
-----
Total time = 22.0017s    Reduced Chi^2 = 1.5637    Levenberg-Marquardt = 0.01
Free parameter values:
    522.2        2.842          115.3          48.32          0.00401
Total time = 22.6107s    Reduced Chi^2 = 1.2669    Levenberg-Marquardt = 0.1
Free parameter values:
    579.6        3.845          235.9          45.52          0.00297
-----
Iteration = 7
-----
Total time = 25.6719s    Reduced Chi^2 = 1.2827    Levenberg-Marquardt = 0.01
Free parameter values:
    614.3        3.324          91.78          56.74          0.005376
Total time = 26.2822s    Reduced Chi^2 = 1.2599    Levenberg-Marquardt = 0.1
Free parameter values:
    588.6        3.694          186.7          49.03          0.003437
-----
Iteration = 8
-----
Total time = 29.3401s    Reduced Chi^2 = 1.2507    Levenberg-Marquardt = 0.01
Free parameter values:
    572.5        3.064          38.61          59.76          0.006845
-----
Iteration = 9
-----
Total time = 32.4461s    Reduced Chi^2 = 1.622     Levenberg-Marquardt = 0.001
Free parameter values:
    682.6        2.691          -370           106.2          0.01729
Total time = 33.067s     Reduced Chi^2 = 1.1959    Levenberg-Marquardt = 0.01
Free parameter values:
    573.2        2.866          -115.3          75.48          0.01053
-----
Iteration = 10
-----
Total time = 36.1542s    Reduced Chi^2 = 1.3469    Levenberg-Marquardt = 0.001
Free parameter values:
    795.4        3.256          -526.2          128.8          0.01636

```

```

Total time = 36.8568s    Reduced Chi^2 = 1.1821    Levenberg-Marquardt = 0.01
Free parameter values:
    601.9      2.849      -281.5      93.91      0.01266
-----
Iteration = 11
-----
Total time = 39.8488s    Reduced Chi^2 = 1.1554    Levenberg-Marquardt = 0.001
Free parameter values:
    497.6      2.384      -504.4      112.7      0.01588
-----
Iteration = 12
-----
Total time = 42.9143s    Reduced Chi^2 = 1.6274    Levenberg-Marquardt = 0.0001
Free parameter values:
    1547      6.038      -1399      263.2      0.01565
Total time = 43.5973s    Reduced Chi^2 = 1.1666    Levenberg-Marquardt = 0.001
Free parameter values:
    814.1      3.54      -795.7      160.5      0.0156
Total time = 44.2453s    Reduced Chi^2 = 1.1467    Levenberg-Marquardt = 0.01
Free parameter values:
    585      2.758      -601.7      127.9      0.01554
-----
Iteration = 13
-----
Total time = 47.3213s    Reduced Chi^2 = 1.1485    Levenberg-Marquardt = 0.001
Free parameter values:
    676.8      3.056      -674.6      140.1      0.01556
Total time = 48.0267s    Reduced Chi^2 = 1.1465    Levenberg-Marquardt = 0.01
Free parameter values:
    613.7      2.851      -625.5      131.7      0.01555
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

Foreground parameter values:
-----
    1          1
    2          8
    3      613.7 +/- 2.053e+04
    4      2.851 +/- 64.84
    5     -625.5 +/- 1.524e+04
    6      131.7 +/- 2620
    7          0
    8          0
    9          0
   10          0

Background parameter values:
-----
    1      0.01555 +/- 0.003024
    2          0
    3          0

Covariance matrix for free parameters:
-----
    1.0000    1.0000   -0.9997    0.9999    0.1227
    1.0000    1.0000   -0.9997    0.9999    0.1220
   -0.9997   -0.9997    1.0000   -1.0000   -0.1439
    0.9999    0.9999   -1.0000    1.0000    0.1365
    0.1227    0.1220   -0.1439    0.1365    1.0000

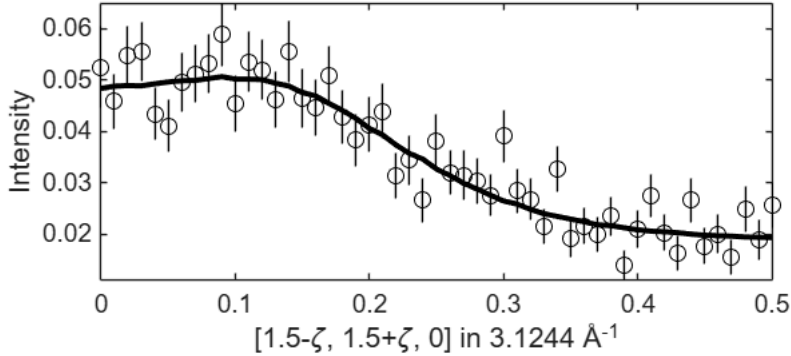
```

Fe_ei800_align_sym3.sqw

i= 2.9± 65; J0=1.3e+02±2.6e+03; J1= 0± 0; gamma=613.667±2053

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $280 \leq$

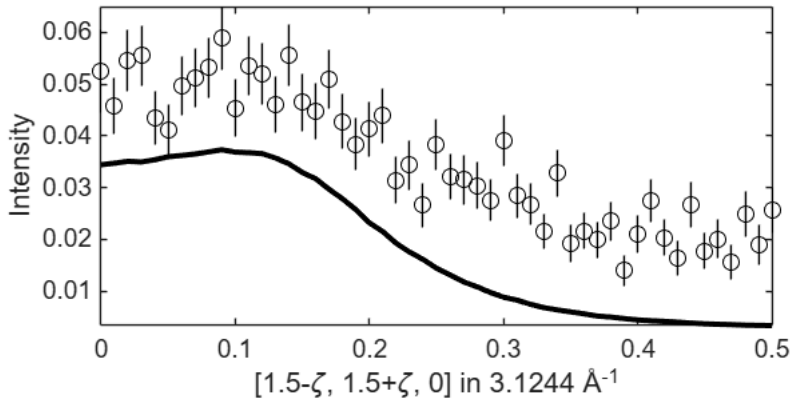
$\zeta=-0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $280 \leq$

$\zeta=-0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



**** En = 295±5 going to 345

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.69528s Reduced Chi^2 = 27.5449

Free parameter values:

613.7	2.851	-625.5	131.7	0.03365
-------	-------	--------	-------	---------

Iteration = 1

Total time = 4.1413s Reduced Chi^2 = 1.7127 Levenberg-Marquardt = 0.1

Free parameter values:

757.6	2.909	-614.1	138.3	0.01591
-------	-------	--------	-------	---------

Iteration = 2

Total time = 7.4484s Reduced Chi^2 = 1.5819 Levenberg-Marquardt = 0.01

Free parameter values:

1424	4.788	-449.2	145.1	0.009975
------	-------	--------	-------	----------

Iteration = 3

Total time = 10.9208s Reduced Chi^2 = 2.1404 Levenberg-Marquardt = 0.001

Free parameter values:

3398	9.274	-174.9	174.7	0.003028
Total time = 11.6154s	Reduced Chi^2 = 1.4899		Levenberg-Marquardt = 0.01	
Free parameter values:				
2229	6.667	-286.6	147.9	0.006318

Iteration = 4

Total time = 15.1435s	Reduced Chi^2 = 2.2148		Levenberg-Marquardt = 0.001	
Free parameter values:				
5138	12.43	-25.21	190.8	-0.0005215
Total time = 15.8192s	Reduced Chi^2 = 1.4607		Levenberg-Marquardt = 0.01	
Free parameter values:				
3265	8.789	-170.8	157.5	0.003116

Iteration = 5

Total time = 19.2233s	Reduced Chi^2 = 1.8372		Levenberg-Marquardt = 0.001	
Free parameter values:				
6523	14.52	115.3	192.6	-0.002564
Total time = 19.9377s	Reduced Chi^2 = 1.4448		Levenberg-Marquardt = 0.01	
Free parameter values:				
4395	10.8	-60.57	166	0.0007868

Iteration = 6

Total time = 23.4674s	Reduced Chi^2 = 1.4491		Levenberg-Marquardt = 0.001	
Free parameter values:				
6178	14.19	247.9	163	-0.004206
Total time = 24.1492s	Reduced Chi^2 = 1.4226		Levenberg-Marquardt = 0.01	
Free parameter values:				
5032	12.01	47.1	163.2	-0.001169

Iteration = 7

Total time = 27.4545s	Reduced Chi^2 = 1.6302		Levenberg-Marquardt = 0.001	
Free parameter values:				
8951	18.18	356.1	195.3	-0.00622
Total time = 28.1366s	Reduced Chi^2 = 1.426		Levenberg-Marquardt = 0.01	
Free parameter values:				
6299	14.07	156.3	172.1	-0.003061
Total time = 28.8164s	Reduced Chi^2 = 1.4126		Levenberg-Marquardt = 0.1	
Free parameter values:				
5435	12.67	79.87	165.9	-0.001783

Iteration = 8

Total time = 32.1892s	Reduced Chi^2 = 1.4337		Levenberg-Marquardt = 0.01	
Free parameter values:				
7429	15.6	198.6	184.8	-0.003731
Total time = 32.8786s	Reduced Chi^2 = 1.4166		Levenberg-Marquardt = 0.1	
Free parameter values:				
6068	13.6	116.8	171.9	-0.002429
Total time = 33.5272s	Reduced Chi^2 = 1.4246		Levenberg-Marquardt = 1	
Free parameter values:				
5638	12.96	91.38	167.8	-0.001989
Total time = 34.2769s	Reduced Chi^2 = 1.4259		Levenberg-Marquardt = 100	
Free parameter values:				
5456	12.7	81	166.1	-0.001803
Total time = 34.9589s	Reduced Chi^2 = 1.421		Levenberg-Marquardt = 10000	
Free parameter values:				
5437	12.67	79.98	165.9	-0.001785
Total time = 35.6987s	Reduced Chi^2 = 1.4248		Levenberg-Marquardt = 1000000	
Free parameter values:				
5436	12.67	79.88	165.9	-0.001783

*** No improvement in chi-squared over previous iteration ***

Total time = 35.7075s Reduced Chi^2 = 1.4126

Free parameter values:

5435 12.67 79.87 165.9 -0.001783

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	5435	+/-	5.494e+04
4	12.67	+/-	62.66
5	79.87	+/-	935.5
6	165.9	+/-	967.5
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	-0.001783	+/-	0.01687
2	0		
3	0		

Covariance matrix for free parameters:

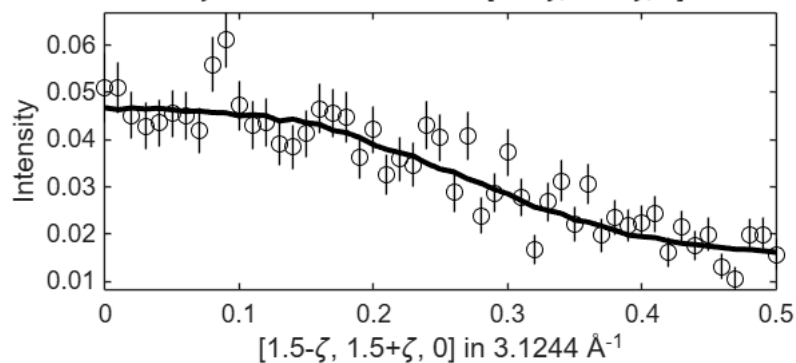
1.0000	0.9999	-0.9884	1.0000	0.9836
0.9999	1.0000	-0.9898	1.0000	0.9818
-0.9884	-0.9898	1.0000	-0.9890	-0.9449
1.0000	1.0000	-0.9890	1.0000	0.9829
0.9836	0.9818	-0.9449	0.9829	1.0000

Fe_ei800_align_sym3.sqw

$\beta = 13 \pm 63$; $J_0 = 1.7e+02 \pm 9.7e+02$; $J_1 = 0 \pm 0$; $\gamma = 5435.3 \pm 5494$

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $290 \leq$

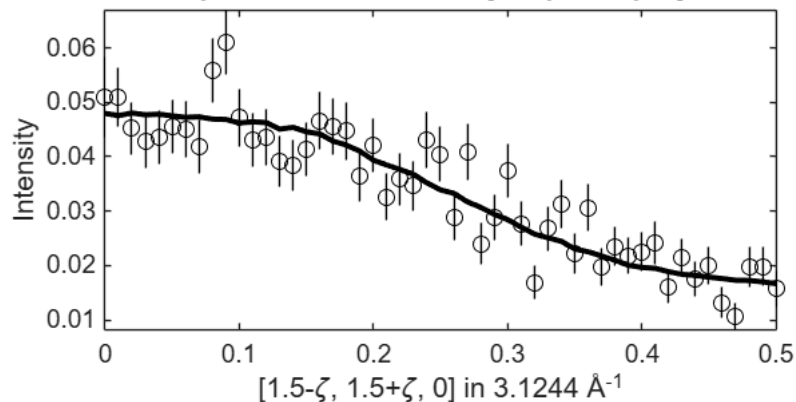
$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $290 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



**** En = 305±5 going to 345

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.82957s Reduced Chi^2 = 105.3629

Free parameter values:

5435 12.67 79.87 165.9 0.03321

Iteration = 1

Total time = 4.9122s Reduced Chi^2 = 3.1956 Levenberg-Marquardt = 0.1

Free parameter values:

5812 9.029 28.71 211.9 0.01512

Iteration = 2

Total time = 8.9408s Reduced Chi^2 = 2.44 Levenberg-Marquardt = 0.01

Free parameter values:

6158 9.749 -1018 269.5 0.005836

Iteration = 3

Total time = 13.0273s Reduced Chi^2 = 1.342 Levenberg-Marquardt = 0.001

Free parameter values:

8567 12.22 -1388 367.6 0.009562

Iteration = 4

Total time = 17.1251s Reduced Chi^2 = 1.4477 Levenberg-Marquardt = 0.0001

Free parameter values:

1.838e+04 19.93 -2666 629.8 0.009758

Total time = 17.9762s Reduced Chi^2 = 1.2786 Levenberg-Marquardt = 0.001

Free parameter values:

1.068e+04 14.29 -1971 457.5 0.009588

Iteration = 5

Total time = 22.0889s Reduced Chi^2 = 1.7252 Levenberg-Marquardt = 0.0001

Free parameter values:

2.679e+04 25.32 -3670 823.6 0.009895

Total time = 22.9169s Reduced Chi^2 = 1.3039 Levenberg-Marquardt = 0.001

Free parameter values:

1.544e+04	17.54	-2575	575.8	0.009848
Total time = 23.7673s		Reduced Chi^2 = 1.2789		Levenberg-Marquardt = 0.01
Free parameter values:				
1.203e+04	15.23	-2194	495.8	0.009671
Total time = 24.609s		Reduced Chi^2 = 1.2822		Levenberg-Marquardt = 1
Free parameter values:				
1.081e+04	14.37	-1998	461.3	0.009548
Total time = 25.5115s		Reduced Chi^2 = 1.2762		Levenberg-Marquardt = 100
Free parameter values:				
1.069e+04	14.29	-1974	457.9	0.009582

Iteration = 6

Total time = 29.4419s		Reduced Chi^2 = 1.2808		Levenberg-Marquardt = 10
Free parameter values:				
1.075e+04	14.33	-1984	459.5	0.00957
Total time = 30.318s		Reduced Chi^2 = 1.283		Levenberg-Marquardt = 100
Free parameter values:				
1.071e+04	14.31	-1977	458.4	0.009578
Total time = 31.1094s		Reduced Chi^2 = 1.2785		Levenberg-Marquardt = 1000
Free parameter values:				
1.07e+04	14.3	-1975	458.1	0.009581
Total time = 31.8713s		Reduced Chi^2 = 1.2788		Levenberg-Marquardt = 100000
Free parameter values:				
1.069e+04	14.3	-1974	457.9	0.009582
Total time = 32.7243s		Reduced Chi^2 = 1.2777		Levenberg-Marquardt = 10000000
Free parameter values:				
1.069e+04	14.29	-1974	457.9	0.009582
Total time = 33.5151s		Reduced Chi^2 = 1.2807		Levenberg-Marquardt = 1000000000
Free parameter values:				
1.069e+04	14.29	-1974	457.9	0.009582

*** No improvement in chi-squared over previous iteration ***

Total time = 33.5236s		Reduced Chi^2 = 1.2762		
Free parameter values:				
1.069e+04	14.29	-1974	457.9	0.009582

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	1.069e+04	+/-	4934
4	14.29	+/-	4.097
5	-1974	+/-	659.1
6	457.9	+/-	17.52
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.009582	+/-	0.003874
2	0		
3	0		

Covariance matrix for free parameters:

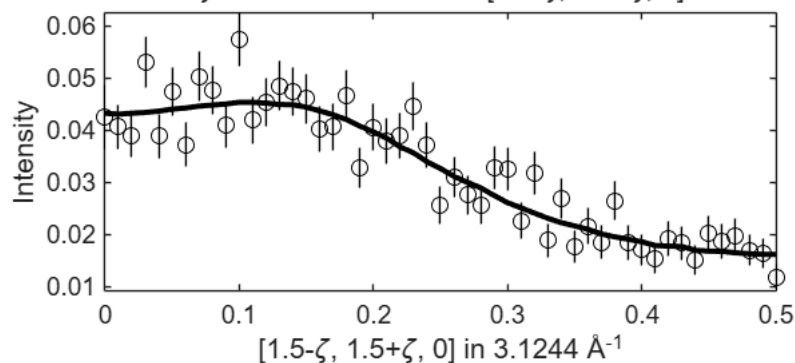
1.0000	0.9795	0.9778	-0.8660	-0.9459
0.9795	1.0000	0.9295	-0.7861	-0.9843
0.9778	0.9295	1.0000	-0.9220	-0.8616
-0.8660	-0.7861	-0.9220	1.0000	0.6898
-0.9459	-0.9843	-0.8616	0.6898	1.0000

Fe_ei800_align_sym3.sqw

S= 14± 4.1; J0=4.6e+02± 18; J1= 0± 0; gamma=10692.2±4933.9

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $300 \leq$

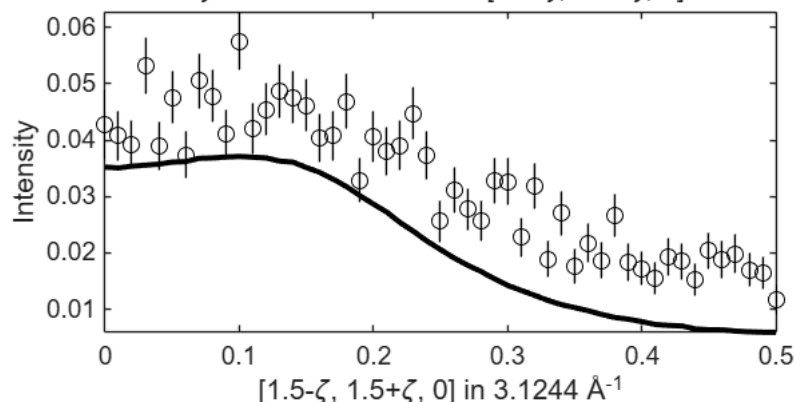
$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $300 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



**** En = 315±5 going to 345

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.83404s Reduced Chi^2 = 41.9215

Free parameter values:

1.069e+04	14.29	-1974	457.9	0.03189
-----------	-------	-------	-------	---------

Iteration = 1

Total time = 5.2941s Reduced Chi^2 = 1.2549 Levenberg-Marquardt = 0.1

Free parameter values:

1.148e+04	11.47	-2311	502.8	0.01546
-----------	-------	-------	-------	---------

Iteration = 2

Total time = 9.9697s Reduced Chi^2 = 1.0655 Levenberg-Marquardt = 0.01

Free parameter values:

1.631e+04	15.48	-2744	580.7	0.01149
-----------	-------	-------	-------	---------

Iteration = 3

Total time = 14.4302s	Reduced Chi^2 = 1.0863	Levenberg-Marquardt = 0.001
Free parameter values:		
2.595e+04	20.77 -3117	698.7 0.01014
Total time = 15.3485s	Reduced Chi^2 = 1.0467	Levenberg-Marquardt = 0.01
Free parameter values:		
1.987e+04	17.68 -2804	612.4 0.0107

Iteration = 4

Total time = 19.5691s	Reduced Chi^2 = 1.2163	Levenberg-Marquardt = 0.001
Free parameter values:		
3.988e+04	26.97 -3958	889.9 0.009896
Total time = 20.3973s	Reduced Chi^2 = 1.053	Levenberg-Marquardt = 0.01
Free parameter values:		
2.659e+04	20.86 -3129	697.3 0.01024
Total time = 21.2683s	Reduced Chi^2 = 1.0542	Levenberg-Marquardt = 0.1
Free parameter values:		
2.206e+04	18.71 -2909	638.5 0.01053
Total time = 22.1126s	Reduced Chi^2 = 1.0451	Levenberg-Marquardt = 10
Free parameter values:		
2.011e+04	17.79 -2817	614.7 0.01069

Iteration = 5

Total time = 26.4007s	Reduced Chi^2 = 1.0452	Levenberg-Marquardt = 1
Free parameter values:		
2.077e+04	18.1 -2845	621.9 0.01064
Total time = 27.3296s	Reduced Chi^2 = 1.0536	Levenberg-Marquardt = 10
Free parameter values:		
2.033e+04	17.88 -2827	616.9 0.01068
Total time = 28.1761s	Reduced Chi^2 = 1.0409	Levenberg-Marquardt = 100
Free parameter values:		
2.018e+04	17.82 -2820	615.4 0.01068

Iteration = 6

Total time = 32.4528s	Reduced Chi^2 = 1.0483	Levenberg-Marquardt = 10
Free parameter values:		
2.042e+04	17.93 -2832	618.1 0.01067
Total time = 33.3327s	Reduced Chi^2 = 1.0492	Levenberg-Marquardt = 100
Free parameter values:		
2.026e+04	17.85 -2824	616.2 0.01068
Total time = 34.2202s	Reduced Chi^2 = 1.0434	Levenberg-Marquardt = 1000
Free parameter values:		
2.021e+04	17.83 -2821	615.7 0.01068
Total time = 35.0732s	Reduced Chi^2 = 1.0503	Levenberg-Marquardt = 100000
Free parameter values:		
2.018e+04	17.82 -2820	615.4 0.01068
Total time = 35.93s	Reduced Chi^2 = 1.0492	Levenberg-Marquardt = 10000000
Free parameter values:		
2.018e+04	17.82 -2820	615.4 0.01068
Total time = 36.8982s	Reduced Chi^2 = 1.0481	Levenberg-Marquardt = 1000000000
Free parameter values:		
2.018e+04	17.82 -2820	615.4 0.01068
*** No improvement in chi-squared over previous iteration ***		
Total time = 36.9073s	Reduced Chi^2 = 1.0409	
Free parameter values:		
2.018e+04	17.82 -2820	615.4 0.01068

Fit converged:

Parameter values (free parameters with error estimates):

 Foreground parameter values:

1	1		
2	8		
3	2.018e+04	+/-	8045
4	17.82	+/-	4.858
5	-2820	+/-	813.8
6	615.4	+/-	26.79
7	0		
8	0		
9	0		
10	0		

 Background parameter values:

1	0.01068	+/-	0.003842
2	0		
3	0		

 Covariance matrix for free parameters:

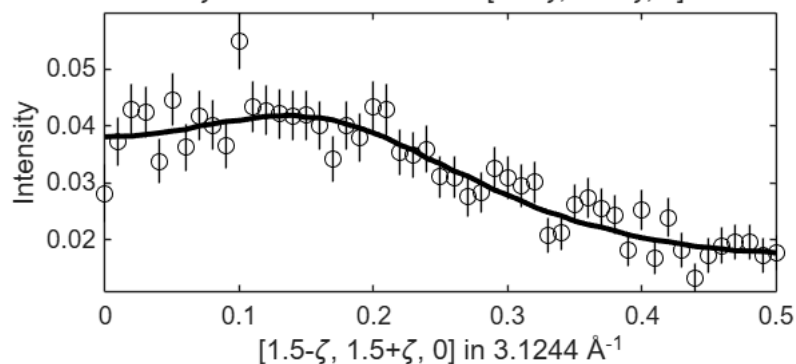
1.0000	0.9710	0.9828	-0.9371	-0.9432
0.9710	1.0000	0.9234	-0.8648	-0.9893
0.9828	0.9234	1.0000	-0.9626	-0.8711
-0.9371	-0.8648	-0.9626	1.0000	0.8015
-0.9432	-0.9893	-0.8711	0.8015	1.0000

Fe_ei800_align_sym3.sqw

S= 18± 4.9; J0=6.2e+02± 27; J1= 0± 0; gamma=20181.6±8045.1

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $310 \leq$

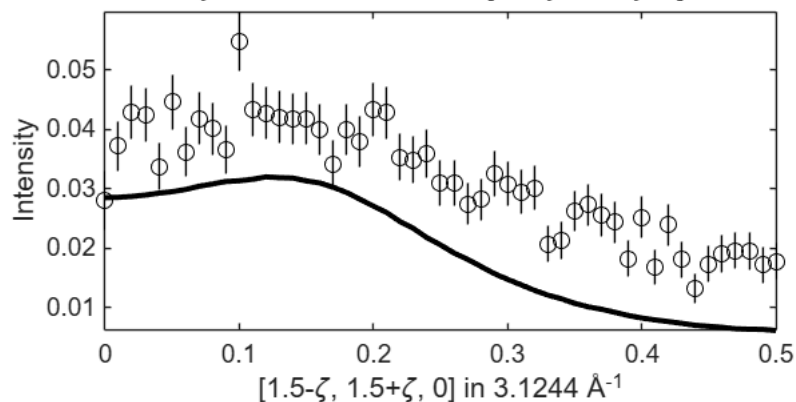
$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $310 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



**** En = 325±5 going to 345

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.75063s Reduced Chi^2 = 28.5437

Free parameter values:

2.018e+04	17.82	-2820	615.4	0.02649
-----------	-------	-------	-------	---------

Iteration = 1

Total time = 4.6092s Reduced Chi^2 = 1.4209

Levenberg-Marquardt = 0.1

Free parameter values:

2.022e+04	12.38	-3029	674.6	0.01564
-----------	-------	-------	-------	---------

Iteration = 2

Total time = 8.3525s Reduced Chi^2 = 1.2542

Levenberg-Marquardt = 0.01

Free parameter values:

2.193e+04	14.52	-3702	751.1	0.01348
-----------	-------	-------	-------	---------

Iteration = 3

Total time = 12.12s Reduced Chi^2 = 1.254

Levenberg-Marquardt = 0.001

Free parameter values:

3.214e+04	18.12	-4497	918.9	0.01337
-----------	-------	-------	-------	---------

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	3.214e+04	+/-	4.53e+07
4	18.12	+/-	1.289e+04
5	-4497	+/-	3.204e+06
6	918.9	+/-	6.509e+05
7	0		
8	0		
9	0		

Background parameter values:

```
-----
1      0.01337  +/-  0.003437
2      0
3      0
```

Covariance matrix for free parameters:

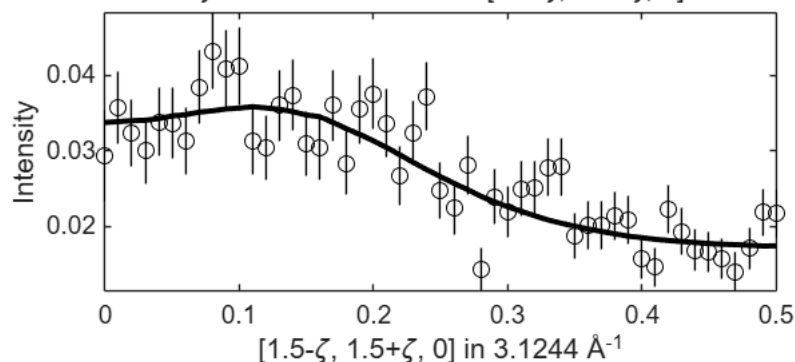
```
-----
1.0000  1.0000  -1.0000  1.0000  0.2348
1.0000  1.0000  -1.0000  1.0000  0.2347
-1.0000 -1.0000  1.0000  -1.0000 -0.2356
1.0000  1.0000  -1.0000  1.0000  0.2352
0.2348  0.2347  -0.2356  0.2352  1.0000
```

Fe_ei800_align_sym3.sqw

$\lambda \pm 1.3e+04$; $J_0 = 9.2e+02 \pm 6.5e+05$; $J_1 = 0 \pm 0$; $\gamma = 32144.8 \pm 4.52$

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $320 \leq$

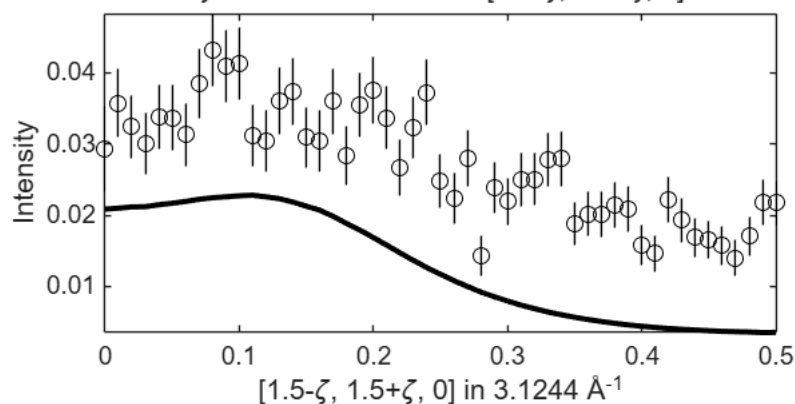
$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $320 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



*** En = 335±5 going to 345

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.62621s Reduced Chi^2 = 10.3246

Free parameter values:

3.214e+04 18.12 -4497 918.9 0.02516

```

-----
Iteration = 1
-----
Total time = 3.7506s    Reduced Chi^2 = 0.99688    Levenberg-Marquardt = 0.1
Free parameter values:
  3.587e+04          16.48          -5041          1021          0.01587
-----
Iteration = 2
-----
Total time = 6.8687s    Reduced Chi^2 = 0.94939    Levenberg-Marquardt = 0.01
Free parameter values:
  5.6e+04           22.8           -4994          1121          0.01328
-----
Iteration = 3
-----
Total time = 9.9703s    Reduced Chi^2 = 1.0494    Levenberg-Marquardt = 0.001
Free parameter values:
  1.239e+05         37.92           -6254          1589          0.01172
Total time = 10.583s    Reduced Chi^2 = 0.94372    Levenberg-Marquardt = 0.01
Free parameter values:
  8.175e+04         28.85           -5145          1250          0.01234
-----
Iteration = 4
-----
Total time = 13.7233s    Reduced Chi^2 = 1.0941    Levenberg-Marquardt = 0.001
Free parameter values:
  1.971e+05         50.02           -7468          1976          0.01131
Total time = 14.3298s    Reduced Chi^2 = 0.94678    Levenberg-Marquardt = 0.01
Free parameter values:
  1.223e+05         36.31           -5702          1467          0.0118
Total time = 14.9346s    Reduced Chi^2 = 0.93876    Levenberg-Marquardt = 0.1
Free parameter values:
  9.58e+04          31.27           -5340          1314          0.01218
-----
Iteration = 5
-----
Total time = 17.9755s    Reduced Chi^2 = 0.9427    Levenberg-Marquardt = 0.01
Free parameter values:
  1.275e+05         36.94           -5525          1453          0.01162
Total time = 18.5946s    Reduced Chi^2 = 0.93489    Levenberg-Marquardt = 0.1
Free parameter values:
  1.064e+05         33.11           -5390          1357          0.01199
-----
Iteration = 6
-----
Total time = 21.6378s    Reduced Chi^2 = 0.93623    Levenberg-Marquardt = 0.01
Free parameter values:
  1.421e+05         39.26           -5689          1518          0.01147
Total time = 22.2757s    Reduced Chi^2 = 0.93866    Levenberg-Marquardt = 0.1
Free parameter values:
  1.18e+05          35.12           -5476          1407          0.0118
Total time = 22.8921s    Reduced Chi^2 = 0.93893    Levenberg-Marquardt = 1
Free parameter values:
  1.101e+05         33.75           -5420          1372          0.01193
Total time = 23.529s     Reduced Chi^2 = 0.93503    Levenberg-Marquardt = 100
Free parameter values:
  1.068e+05         33.18           -5393          1358          0.01198
Total time = 24.1443s    Reduced Chi^2 = 0.93744    Levenberg-Marquardt = 10000
Free parameter values:
  1.064e+05         33.12           -5390          1357          0.01199
Total time = 24.7765s    Reduced Chi^2 = 0.93796    Levenberg-Marquardt = 1000000
Free parameter values:
  1.064e+05         33.11           -5390          1357          0.01199
*** No improvement in chi-squared over previous iteration ***

```

Total time = 24.7848s Reduced Chi^2 = 0.93489
 Free parameter values:
 1.064e+05 33.11 -5390 1357 0.01199

 Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	1.064e+05	+/-	7.442e+04
4	33.11	+/-	16.8
5	-5390	+/-	3899
6	1357	+/-	138.7
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.01199	+/-	0.004778
2	0		
3	0		

Covariance matrix for free parameters:

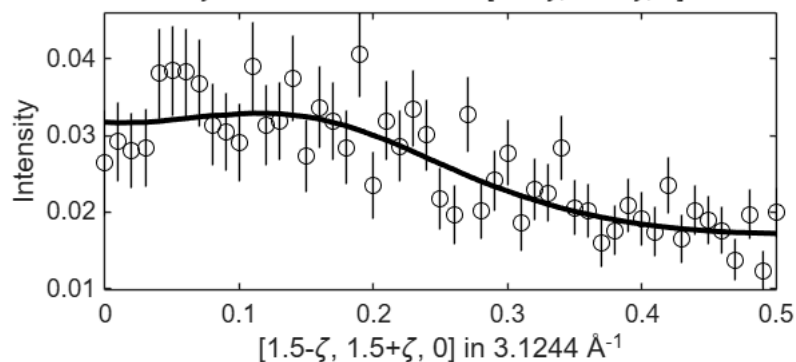
1.0000	0.9790	0.9952	-0.9822	-0.9428
0.9790	1.0000	0.9680	-0.9788	-0.9831
0.9952	0.9680	1.0000	-0.9688	-0.9152
-0.9822	-0.9788	-0.9688	1.0000	0.9547
-0.9428	-0.9831	-0.9152	0.9547	1.0000

Fe_ei800_align_sym3.sqw

$\lambda = 33 \pm 17$; $J_0 = 1.4 \times 10^3 \pm 1.4 \times 10^2$; $J_1 = 0 \pm 0$; $\gamma = 106409 \pm 7442$

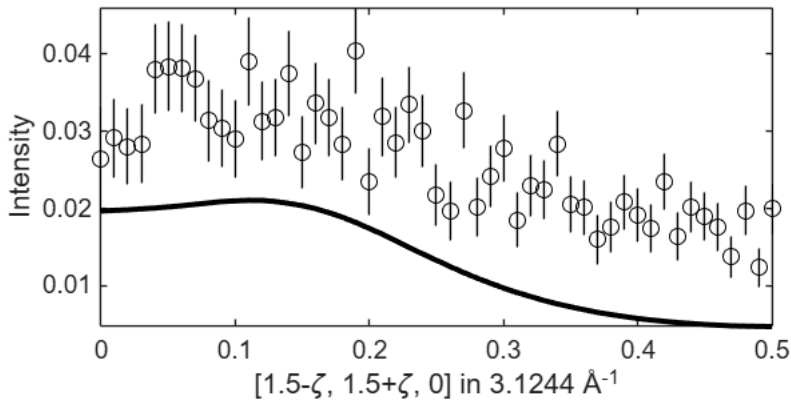
$\xi \leq 0.1$ in $[1.5 + \xi, 1.5 + \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $330 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5 - \zeta, 1.5 + \zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5+\xi, 1.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $330 \leq \zeta = -0.005:0.01:0.505$ in $[1.5-\zeta, 1.5+\zeta, 0]$



 **** En = 345±5 going to 345

Beginning fit (max 20 iterations)-----
 Starting point

Total time = 0.26212s Reduced Chi^2 = 7.6696
 Free parameter values:
 1.064e+05 33.11 -5390 1357 0.01896

Iteration = 1

Total time = 1.6299s Reduced Chi^2 = 1.4151 Levenberg-Marquardt = 0.1
 Free parameter values:
 8.369e+04 24.43 -6910 1481 0.01053

Iteration = 2

Total time = 3.0363s Reduced Chi^2 = 1.5422 Levenberg-Marquardt = 0.01
 Free parameter values:
 4.021e+04 22.3 -3791 834.1 0.007448
 Total time = 3.2975s Reduced Chi^2 = 1.3728 Levenberg-Marquardt = 0.1
 Free parameter values:
 6.569e+04 24.8 -6475 1256 0.008547

Iteration = 3

Total time = 4.6716s Reduced Chi^2 = 3.2479 Levenberg-Marquardt = 0.01
 Free parameter values:
 3211 14.34 -3349 662.4 0.007803
 Total time = 4.9687s Reduced Chi^2 = 1.3646 Levenberg-Marquardt = 0.1
 Free parameter values:
 4.369e+04 21.68 -5516 1070 0.00805

Iteration = 4

Total time = 6.3326s Reduced Chi^2 = 1.4171 Levenberg-Marquardt = 0.01
 Free parameter values:
 1.637e+04 14.54 -3290 667.4 0.007634
 Total time = 6.6501s Reduced Chi^2 = 1.3565 Levenberg-Marquardt = 0.1
 Free parameter values:
 3.6e+04 19.36 -4830 940.4 0.007868

Iteration = 5

```

-----
Total time = 8.0245s   Reduced Chi^2 = 1.4566   Levenberg-Marquardt = 0.01
Free parameter values:
  1.263e+04   13.16   -3165   623.2   0.007699
Total time = 8.3125s   Reduced Chi^2 = 1.3686   Levenberg-Marquardt = 0.1
Free parameter values:
  2.858e+04   17.44   -4310   839.7   0.007779
Total time = 8.6035s   Reduced Chi^2 = 1.3627   Levenberg-Marquardt = 1
Free parameter values:
  3.364e+04   18.76   -4668   908.4   0.007832
Total time = 8.879s   Reduced Chi^2 = 1.3635   Levenberg-Marquardt = 100
Free parameter values:
  3.577e+04   19.3   -4813   937.2   0.007864
Total time = 9.1478s   Reduced Chi^2 = 1.3594   Levenberg-Marquardt = 10000
Free parameter values:
  3.598e+04   19.36   -4828   940   0.007868
Total time = 9.4332s   Reduced Chi^2 = 1.3652   Levenberg-Marquardt = 1000000
Free parameter values:
  3.6e+04   19.36   -4829   940.3   0.007868
*** No improvement in chi-squared over previous iteration ***
Total time = 9.4426s   Reduced Chi^2 = 1.3565
Free parameter values:
  3.6e+04   19.36   -4830   940.4   0.007868
-----

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
 1          1
 2          8
 3      3.6e+04 +/-  3.236e+04
 4      19.36 +/-  11.04
 5      -4830 +/-  3222
 6      940.4 +/-  149.5
 7          0
 8          0
 9          0
10          0

```

Background parameter values:

```

-----
 1      0.007868 +/-  0.005244
 2          0
 3          0

```

Covariance matrix for free parameters:

```

-----
 1.0000   0.9384   0.9871  -0.9838  -0.9280
 0.9384   1.0000   0.9126  -0.9322  -0.9847
 0.9871   0.9126   1.0000  -0.9794  -0.8767
-0.9838  -0.9322  -0.9794   1.0000   0.9049
-0.9280  -0.9847  -0.8767   0.9049   1.0000

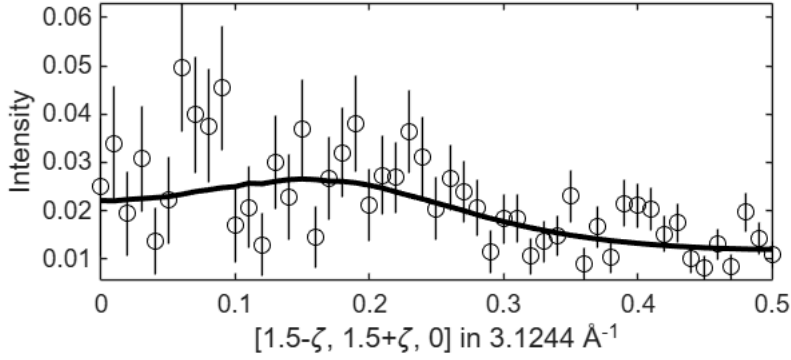
```

Fe_ei800_align_sym3.sqw

$i = 19 \pm 11$; $J_0 = 9.4 \times 10^2 \pm 1.5 \times 10^2$; $J_1 = 0 \pm 0$; $\gamma = 36002.4 \pm 3235$

$\xi \leq 0.1$ in $[1.5 + \xi, 1.5 + \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $340 \leq$

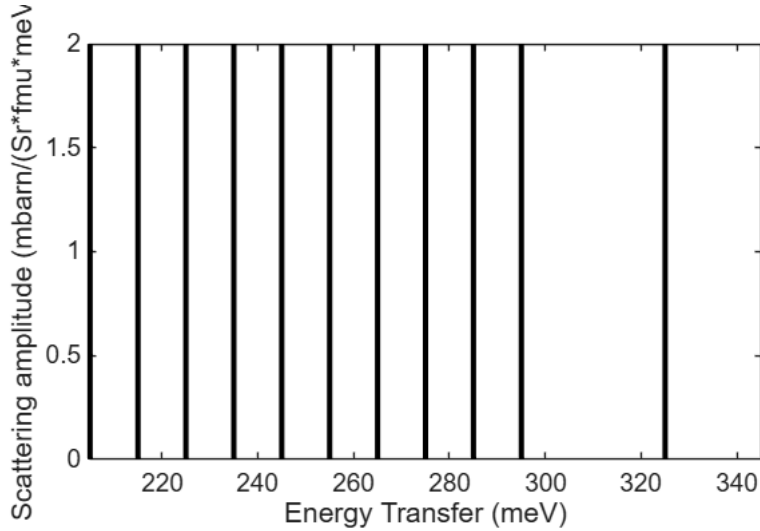
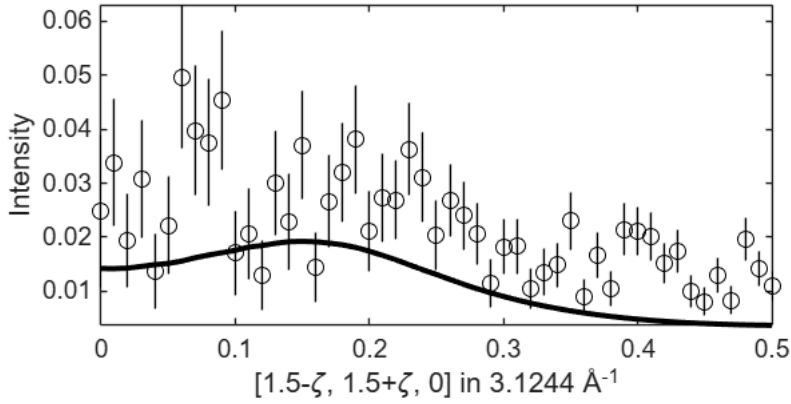
$\zeta = -0.005:0.01:0.505$ in $[1.5 - \zeta, 1.5 + \zeta, 0]$

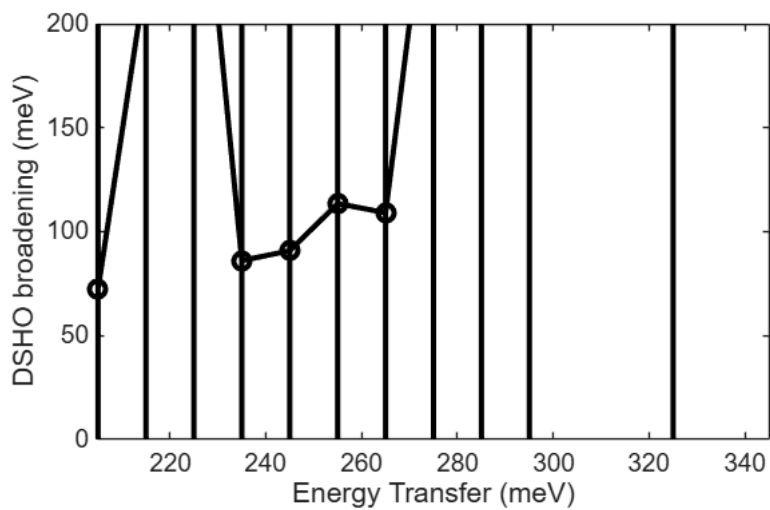
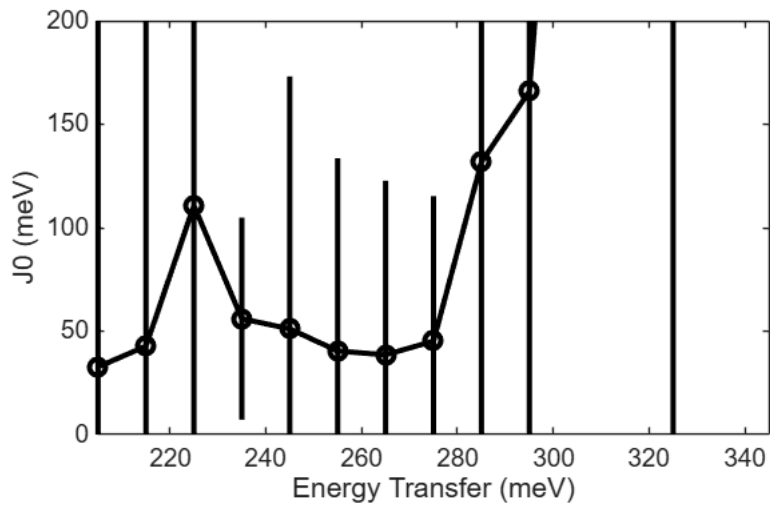


Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[1.5 + \xi, 1.5 + \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1.5, 1.5, -\eta]$, $340 \leq$

$\zeta = -0.005:0.01:0.505$ in $[1.5 - \zeta, 1.5 + \zeta, 0]$

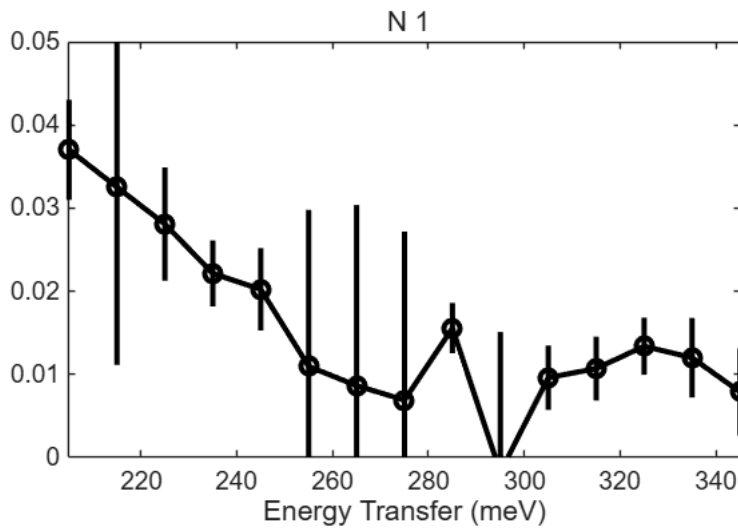




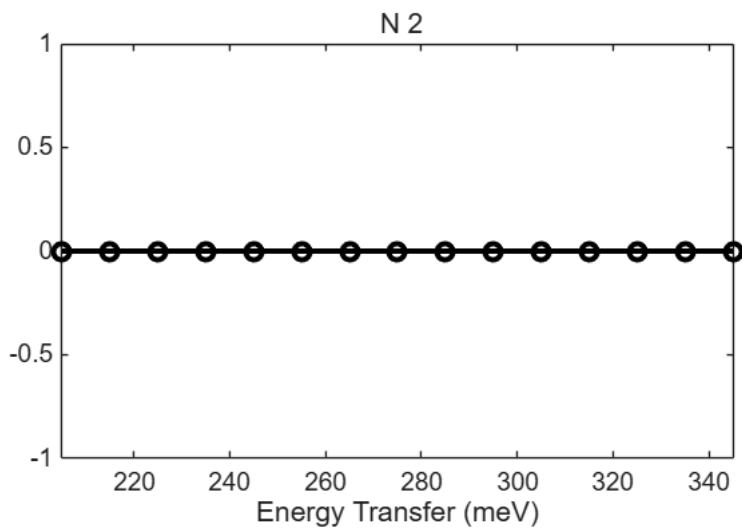
```
c1 = HN_FFCutsFits.(selectCuts);
HN_FFCutsFits.(selectCuts) = [c1{:}];
```

```
beep
beep
beep
```

```
all_fit_par = HN_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);ly 0 0.05; keep_figure;
```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_HN.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_HN_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struct = HN_FFCutsFits;
save(CutFitFileHN,"fit_src_struct","-v7.3')
```